Ofqual QAN	**600/3055/0**
Quartz ID	**1158**
Qualification	**Level 5 Diploma in Computing**

Assessment	**Network Security and Cryptography 2.0**
Assessment Type	**Examination**
Date	**Wednesday 5th March 2025**
Time	**09.00am – 12.00pm**

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

This marking scheme has been prepared as a **guide only** to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Marker's comments:

Moderator's comments:

Mark:	**Moderated mark:**	**Final mark:**

Penalties applied for academic malpractice:

Answer ALL questions

Marks

Question 1

a) Describe the Caesar cipher and encrypt the word **SECURITY** using a Caesar cipher with a key of **FIVE (5)**. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Caesar ciphers are among the simplest devised and rely solely on remapping characters to others in the alphabet using a constant shift modulo the size of the alphabet (1 mark).***
- ***The amount shifted is the key used to encipher, or decipher, the message. (1 mark)***
- ***Shift each letter in SECURITY by 5 places to the right: XJHZWNYD (1 mark)***

S	***E***	***C***	***U***	***R***	***I***	***T***	***Y***
X	***J***	***H***	***Z***	***W***	***N***	***Y***	***D***

(1 mark for showing workings)

b) Explain the **THREE (3)** core principles of network security in relation to the CIA triad. **6**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 6 marks:

- ***Confidentiality (1 mark): Ensures that information is only accessible to those who are authorised to view it. (1 mark)***
- ***Integrity (1 mark): Ensures that information is accurate and has not been tampered with or altered by unauthorised parties. (1 mark)***
- ***Availability (1 mark): Ensures that information and resources are accessible to authorised users when needed. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 2

a) Describe **THREE (3)** methods for securely sharing a secret key used by two parties in symmetric cryptography. **6**

Mark Scheme

Award 1 mark for any of the following up to a maximum of 6 marks:

- ***Pre-Shared Key (PSK) (1 mark): Both parties agree on a secret key in advance before communication begins. (1 mark)***
- ***Key Distribution Centre (KDC) (1 mark): A trusted third party facilitates the secure distribution of the secret key to both parties. (1 mark)***
- ***Diffie-Hellman Key Exchange (1 mark): Enables two parties to establish a shared secret key over an insecure channel without transmitting the key directly. (1 mark)***
- ***Public Key Infrastructure (PKI) (1 mark): Uses public and private key pairs to securely distribute symmetric keys for encryption. (1 mark)***

b) Explain the difference between key length and entropy **and** explain how each contributes to ensuring cryptographic security. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***The key length refers to the number of bits in the secret key used for encryption (1 mark). Longer key lengths provide stronger security against brute-force attacks (1 mark)***
- ***Entropy is a measure of the randomness or unpredictability of the secret key (1 mark). Higher entropy means the key has more randomness, making it harder for an attacker to guess or predict the key. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 3

a) Explain the primary purpose of a Message Authentication Code (MAC) **and** explain the steps involved in generating and verifying a MAC. **6**

Mark Scheme

Award 2 marks for any of the following up to a maximum of 6 marks:

- ***A Message Authentication Code (MAC) is a cryptographic technique used to ensure the authenticity (1 mark) and integrity of a message (1 mark).***
- ***The sender uses a secret key (1 mark) and a MAC algorithm to generate a MAC value (also called a tag) based on the message content. (1 mark).***
- ***The receiver computes their own MAC value for the received message using the same key and algorithm. (1 mark).***
- ***If the computed MAC matches the received MAC, the message is considered authentic. (1 mark).***

b) An employee receives an email from a colleague with an attachment that claims to be an urgent report. However, the email address seems slightly different from the colleague's usual email address. **4**

Identify the type of attack being described **and** suggest **THREE (3)** methods to prevent it from being successful.

Mark Scheme

Award up to 1 mark for each of the following up to 4 marks:

- ***The attack is a phishing attack (1 mark), specifically a type of email spoofing. (1 mark)***
- ***Use email filtering software or services that detect and block phishing emails. (1 mark)***
- ***Contact the sender directly through another verified method. (1 mark)***
- ***Employee awareness and training to detect phishing attacks. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 4

a) Explain the concept of a format string vulnerability **and** discuss **TWO (2)** potential risks associated with this type of vulnerability. **6**

Mark Scheme

Award 2 marks for definition:

- ***A format string vulnerability occurs when user input is improperly handled in functions that process formatted output (1 mark).***
- ***If user input is directly used as the format string without validation, an attacker can exploit it to execute unintended commands or access restricted areas of memory. (1 mark).***

Award 1 mark for any of the following points to a maximum of 4 marks:

- ***Arbitrary Code Execution (1 mark): If an attacker can control the format string, they might be able to execute arbitrary code on the system. (1 mark).***
- ***Memory Corruption (1 mark): Malicious format strings can corrupt memory, which can lead to crashes or the execution of attacker-controlled code. (1 mark).***
- ***Information Disclosure (1 mark): Attackers can use format strings to read out sensitive information from the memory, such as passwords or encryption keys. (1 mark).***
- ***Denial of Service (DoS) (1 mark): By causing the program to crash or consume excessive resources, attackers can disrupt services, leading to a denial of service. (1 mark).***

b) Discuss **TWO (2)** advantages of biometric authentication compared to password-based authentication. **4**

Mark Scheme

Award 1 mark for any of the following to a maximum of 4 marks:

- ***Unique Identification (1 mark): Biometrics are unique to an individual, such as fingerprints, iris patterns or facial features. This uniqueness enhances the accuracy of identification. (1 mark)***
- ***Difficult to Forge (1 mark): Biometric data is difficult to replicate or forge, making it more secure than traditional passwords. For example, duplicating someone's fingerprint or replicating their iris pattern is a highly complex task. (1 mark)***
- ***Convenience (1 mark): Biometric authentication is often more convenient for users. They don't need to remember complex passwords, as their biological traits serve as their credentials. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 5

a) A web-based login system allows users to input their username and password. The application directly uses these inputs to construct an SQL query to verify credentials. An attacker exploits this by entering a specially crafted malicious SQL query in the username field, which is then stored in the system's database. **4**

- Identify the type of vulnerability present in this scenario.
- Explain **TWO (2)** security mitigations that should be implemented to prevent this vulnerability.

Mark Scheme

Award 1 mark for each up to a maximum of 4 marks:

- ***The vulnerability described is a type of SQL injection (1 mark); typically known as Second-order SQL injection (1 mark).***
- ***Mitigation involves implementing parameterised queries to separate user input from SQL commands (1 mark)***
- ***Mitigation involves input validation to sanitise user input (1 mark)***

b) A company's network has been compromised by ransomware which has spread within its internal network. **6**

Define ransomware **and** discuss **TWO (2)** potential impacts of a ransomware attack on the organisation.

Mark Scheme

Award maximum of 2 marks for definition:

- ***Ransomware is a type of malware designed to deny access to a computer system or its data by encrypting the victim's files (1 mark) until a ransom is paid in exchange for a decryption key (1 mark).***

Award 1 mark for any of the following up to a maximum of 4 marks:

- ***Operational Disruption (1 mark): Ransomware can cripple critical systems, halting business operations and disrupting services. (1 mark)***
- ***Financial Loss (1 mark): Ransomware can lead to direct financial loss, either through theft or the cost of remediation and recovery. (1 mark)***
- ***Reputational Damage (1 mark): The organisation can suffer reputational harm due to loss of customer data or service interruptions. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 6

a) A company aims to secure its email communications to safeguard sensitive information and ensure message authenticity. Recommend and describe **THREE (3)** protocols that can be implemented to achieve these goals. **6**

Mark Scheme

Award 1 mark for each up to a maximum of 6 marks:

- ***PGP (Pretty Good Privacy) (1 mark): enables secure email communication by encrypting and decrypting messages, providing digital signatures for message authentication and encrypting files for secure transmission. (1 mark)***
- ***S/MIME (Secure/Multipurpose Internet Mail Extensions) (1 mark): is an email security protocol that offers encryption and digital signatures for email messages. It enhances the confidentiality and integrity of email communication by encrypting message content and providing a digital signature for message authenticity. (1 mark).***
- ***DKIM (DomainKeys Identified Mail) (1 mark): is an email authentication method that uses digital signatures to verify that an email message was sent and authorised by the owner of a domain. It adds a signature to the email header, allowing the recipient to validate the sender's authenticity. (1 mark).***

b) Explain in detail **TWO (2)** good practices for securing web applications against Cross-Site Scripting (XSS) attacks. **4**

Mark Scheme

Award 1 mark for any of the following up to a maximum of 4 marks:

- ***Input Validation and Sanitisation (1 mark): Validate and sanitise all user inputs to ensure they adhere to expected patterns. (1 mark)***
- ***Content Security Policy (CSP) (1 mark): Implement a Content Security Policy that defines which resources can be loaded and executed on a web page. (1 mark)***
- ***HTTP Secure (HTTPS) (1 mark): Enforce the use of HTTPS to encrypt data in transit between the user's browser and the web server. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 7

a) Explain the difference between signature-based intrusion detection **and** anomaly-based intrusion detection systems. **6**

Describe **ONE (1)** advantage **and ONE (1)** disadvantage of each system.

Mark Scheme

Award 1 mark for each of the following up to a maximum of 6 marks:

- ***Signature-based intrusion detection relies on known patterns of malicious behaviour. (1 mark)***
- ***Anomaly-based intrusion detection looks for deviations from normal system behaviour. (1 mark)***
- ***The advantage of signature-based intrusion detection is its accuracy in identifying known threats (1 mark), but it may miss new or unknown attacks. (1 mark)***
- ***The advantage of anomaly-based intrusion detection is that it can identify novel threats (1 mark) but may generate false positives. (1 mark)***

b) What specific elements of a packet does a packet filtering firewall examine to determine whether to allow or block it? **4**

Mark Scheme

Award 1 mark for each of following up to a maximum of 4 marks:

- ***A packet-filtering firewall examines the source IP address (1 mark) and destination IP address (1 mark), port numbers (1 mark) and protocol type (1 mark) within a packet.***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 8

a) Company ABC intends to use VPNs (Virtual Private Networks) to ensure secure communication between remote employees and their several branch networks. **6**

Explain how a VPN ensures secure communication **and** describe **TWO (2)** types of VPNs that you will recommend.

Mark Scheme

Award 1 mark for any of the following up to a maximum of 6 marks:

- ***VPN is a technology that creates a secure, encrypted connection (1 mark) over a public network, typically the Internet. (1 mark)***
- ***Site-to-Site VPN (1 mark) is best suited for secure connectivity between branch networks and the headquarters. (1 mark)***
- ***Remote Access VPN (1 mark) is ideal for secure communication between individual remote employees and the company network. (1 mark)***

b) Explain the difference between a Local Area Network (LAN) **and** a Virtual Local Area Network (VLAN). **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***A virtual LAN (VLAN) is a logical local area network (1 mark) that groups a subset of devices that share a physical LAN, isolating the traffic for each group. (1 mark)***
- ***A LAN is a group of computers or other devices that share the same physical network (1 mark) and are directly connected to the same physical switch. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 9

a) Describe the concept of Wireless LAN for Trusted Systems (WLTS) **and** its applications. **4**

Mark Scheme

Award 1 mark for each up to a maximum of 4 marks:

- ***WLTS is a security architecture designed for wireless networks (1 mark), focusing on securing communication between trusted devices (1 mark). WLTS ensures that only authorised and trusted systems (1 mark) can access the wireless network, enhancing overall network security. (1 mark)***

b) Company XYZ recently purchased and installed various IoT devices such as smart thermostats and security cameras on their network. **6**

Discuss **THREE (3)** measures the company can take to enhance the security of its network to reduce the risk of vulnerabilities associated with IoT devices.

Mark Scheme

Award 1 mark for any of the following up to a maximum of 6 marks:

- ***Change Default Passwords (1 mark): Set unique, strong passwords for each IoT device. (1 mark)***
- ***Regular Updates (1 mark): Keep firmware and software up to date to patch security vulnerabilities. (1 mark)***
- ***Network Segmentation (1 mark): Isolate IoT devices on a separate network to prevent unauthorised access. (1 mark)***
- ***Use Encryption (1 mark): Ensure data transmitted between devices is encrypted to protect privacy. (1 mark)***

Total 10 Marks

© NCC Education Limited 2025

Marks

Question 10

a) Explain the term Risk Management **and** describe the difference between Risk Appetite **and** Risk Tolerance in relation to information security. **6**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 6 marks:

- ***Risk Management: The process of identifying, assessing and prioritising potential risks (1 mark) to an organisation's information assets. (1 mark)***
- ***Risk Appetite: The amount of risk an organisation is willing to accept (1 mark) in pursuit of its objectives. (1 mark)***
- ***Risk Tolerance: The acceptable level of risk or variation of risk (1 mark) an organisation is willing to tolerate in the achievement of its objectives. (1 mark)***

b) Discuss the differences between qualitative **and** quantitative risk assessments in information security. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Qualitative Risk Assessment: Involves subjective evaluation of risks (1 mark), often using descriptive terms like low, medium or high. (1 mark)***
- ***Quantitative Risk Assessment: Involves numerical analysis (1 mark), often assigning monetary values to risks. (1 mark)***

Total 10 Marks

End of Paper

© NCC Education Limited 2025

Learning Outcomes matrix

Question	Learning Outcomes assessed	Markers can differentiate between varying levels of achievement
1	1,2	Yes
2	1,2	Yes
3	2,3,4	Yes
4	4	Yes
5	4	Yes
6	4	Yes
7	4	Yes
8	4	Yes
9	1,5	Yes
10	6	Yes

Grade descriptors

Learning Outcome	Pass	Merit	Distinction
Understand the basic concept and application of Network Security and Cryptography	Demonstrate adequate understanding and application of Network Security and Cryptography	Demonstrate robust understanding and application of Network Security and Cryptography algorithm	Demonstrate a highly comprehensive understanding and application of Network Security and Cryptography.
Critically evaluate the suitability of different Cryptographic Algorithms	Demonstrates a basic evaluation of different cryptographic algorithms, identifying some strengths and weaknesses.	Demonstrates a thorough evaluation of different cryptographic algorithms, with a detailed analysis of their strengths and weaknesses, and appropriate comparisons.	Demonstrates a comprehensive and critical evaluation of different cryptographic algorithms, with in-depth analysis, insightful comparisons, and a well-argued justification of their suitability in various contexts.
Understand the vulnerabilities and systematically evaluate the security risks in networks	Demonstrates a basic understanding of vulnerabilities and can evaluate security risks in networks with guidance.	Demonstrates a good understanding of vulnerabilities and can systematically evaluate security risks in networks with some analysis.	Demonstrates a comprehensive understanding of vulnerabilities and can critically and systematically evaluate security risks in networks with detailed analysis.
Critically analyse appropriate tools and techniques for network security	Demonstrates a basic ability to analyse appropriate tools and techniques for network security, with limited critical evaluation.	Demonstrates a good ability to analyse appropriate tools and techniques for network security, with a clear and reasoned critical evaluation.	Demonstrates an excellent ability to critically analyse appropriate tools and techniques for network security, providing comprehensive and insightful evaluation.

© NCC Education Limited 2025

Understand the wireless network security and associated protocols	Demonstrates a basic understanding of wireless network security and its associated protocols.	Demonstrates a good understanding of wireless network security and its associated protocols, with clear explanations and some examples.	Demonstrates a comprehensive understanding of wireless network security and its associated protocols, providing detailed explanations, examples, and critical insights.
Explain the concepts of Information Security and explain good practices in achieving security.	Demonstrates a basic understanding of the concepts of Information Security and provides examples of good practice in achieving security.	Demonstrates a clear and detailed understanding of the concepts of Information Security, explaining them accurately and providing well-chosen examples of good practices in achieving security.	Demonstrates a comprehensive and insightful understanding of the concepts of Information Security, explaining them thoroughly and providing detailed, relevant examples of best practices in achieving security.

© NCC Education Limited 2025